132/33 kV
DIGITAL SUBSTATION

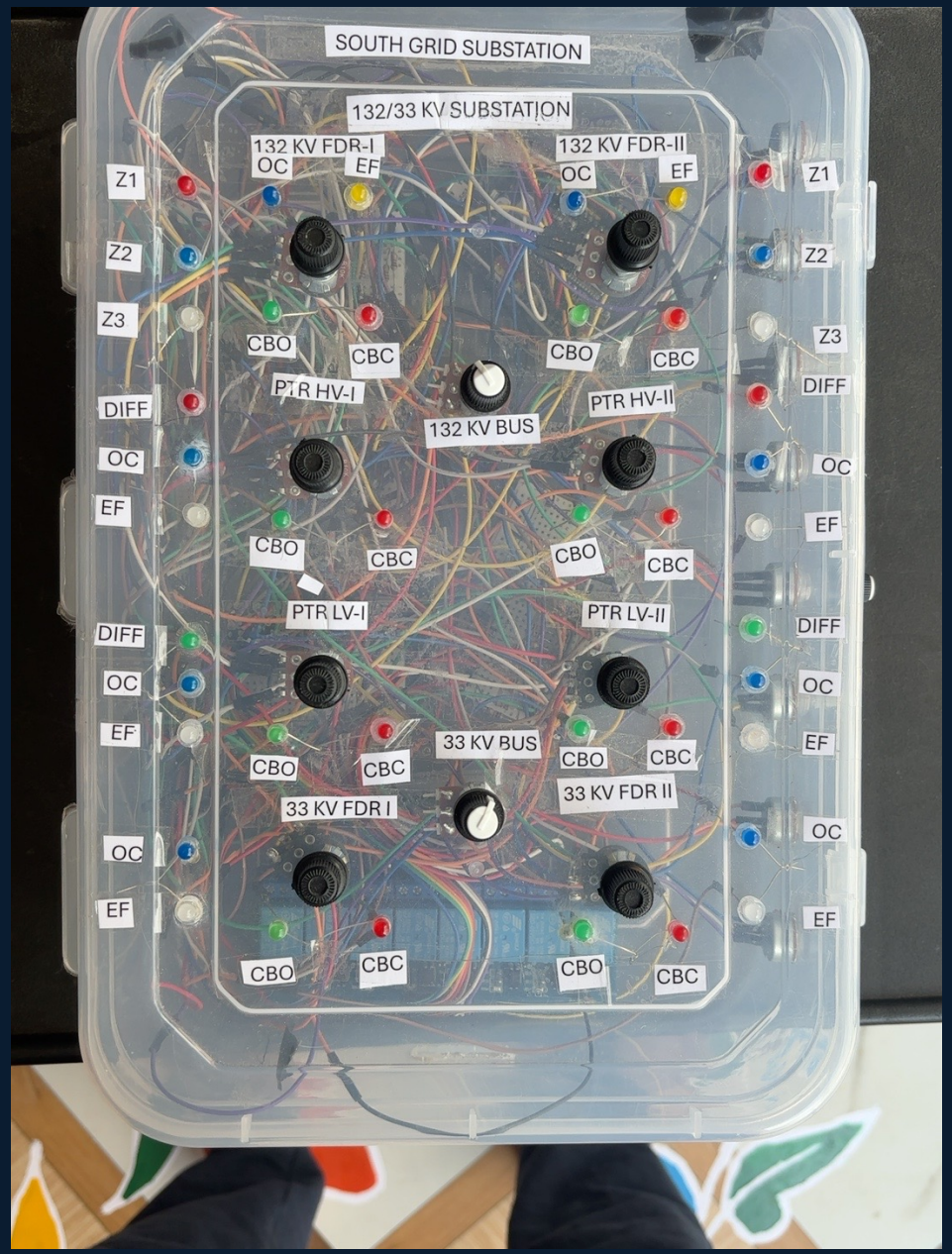
Automation Prototype

ESP32 • Protection • Breaker Control

UART • MQTT • Blynk

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B.Tech – Electrical & Electronics Engineering



Project Overview

A physical low-voltage prototype combining substation mimic, protection logic and IoT supervision.

Measurement

132 kV / 33 kV bus voltage, feeder current and transformer HV/LV current.

Protection

Distance, overcurrent, differential and earth-fault functions.

Automation

Eight relay outputs for prototype breaker operation.

Communication

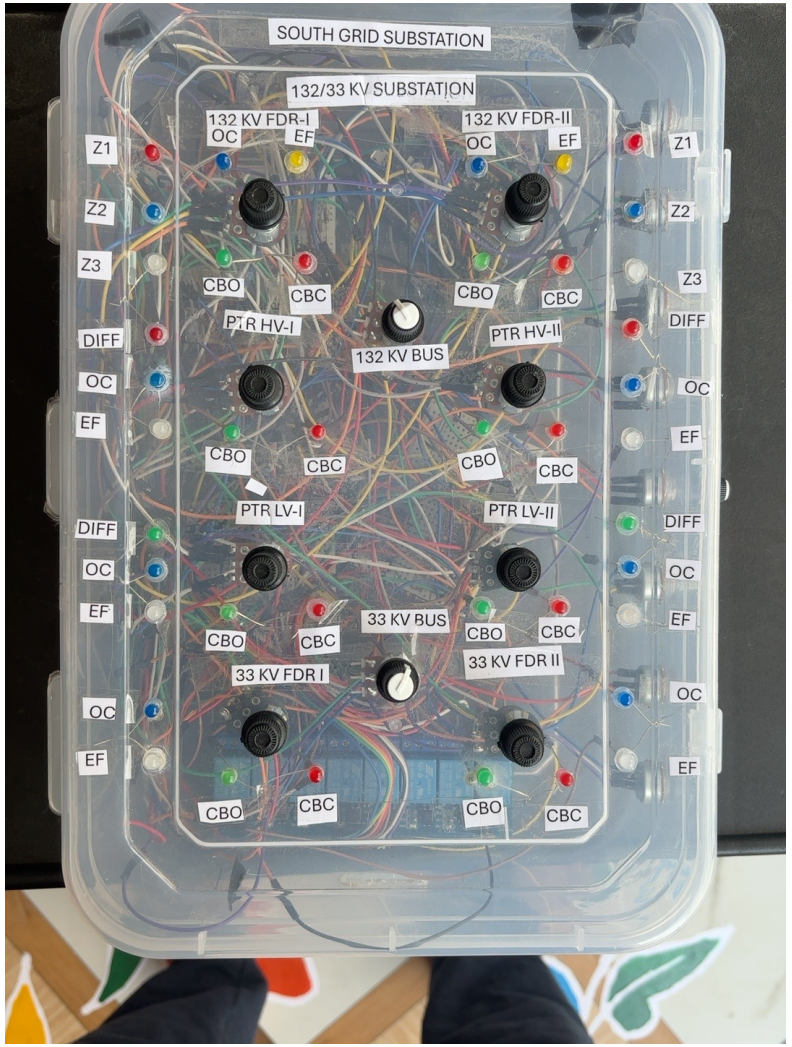
UART event exchange and MQTT event communication.

Supervision

Blynk dashboard for measurements, status and control.

Physical Digital-Substation Prototype

Front-panel mimic and internal controller/relay wiring





Substation Model Represented in the Prototype

132 kV FDR-1

Distance + OC

132 kV FDR-2

Distance + OC

PTR-1

HV/LV + DIFF + OC/EF

PTR-2

HV/LV + DIFF + OC/EF

33 kV FDR-1

OC + EF

33 kV FDR-2

OC + EF

Controller & Communication Architecture

Distributed prototype control with local UART events and cloud MQTT supervision



UART carries protection event strings between the controllers. The ESP32-S3 handles measurement/protection and relay outputs; the ESP32-DEV handles earth-fault processing, annunciation and MQTT messaging.

Blynk IoT

Measurements • status • breaker control

Measurement & Telemetry

Analog inputs are converted to prototype engineering values and published to Blynk.

132 kV bus voltage

Real-time prototype measurement path

33 kV bus voltage

Real-time prototype measurement path

132 kV feeder currents

Real-time prototype measurement path

Transformer HV currents

Real-time prototype measurement path

Transformer LV currents

Real-time prototype measurement path

33 kV feeder currents

Real-time prototype measurement path

Approximate power

The source calculates demonstration power values from measured voltage and current.

Protection Functions Implemented in the Prototype

Distance

Zone 1 • Zone 2 • Zone 3

Overcurrent

132 kV + 33 kV feeder paths

Differential

PTR-1 + PTR-2

Transformer OC

HV OC + LV OC

Earth Fault

Feeders + transformer HV/LV

Distance Protection

Prototype three-zone logic using approximate apparent impedance Z = V / I

Zone 1

Instantaneous Z ≤ 0.264

Zone 2

1 second delay Z ≤ 0.330

Zone 3

2 second delay Z ≤ 0.377

Measured V + I → **Z = V / I** → **Zone decision** → **Trip / annunciation / UART event**

Thresholds and delays shown above are prototype values from the supplied firmware.

Overcurrent & Transformer Protection

Feeder OC

132 kV feeder-1 and feeder-2 33 kV feeder-1 and feeder-2

PTR-1

Differential + HV OC + LV OC + EF

PTR-2

Differential + HV OC + LV OC + EF

Event handling

Protection flags reduce repeated event transmission; reset messages are generated when the prototype condition clears.

Earth-Fault Detection & MQTT Event Gateway

Eight EF channels

132 kV FDR-1/FDR-2 PTR-1 HV/LV PTR-2 HV/LV
33 kV FDR-1/FDR-2

Protection action

Analog pickup → annunciation → UART event → MQTT publication

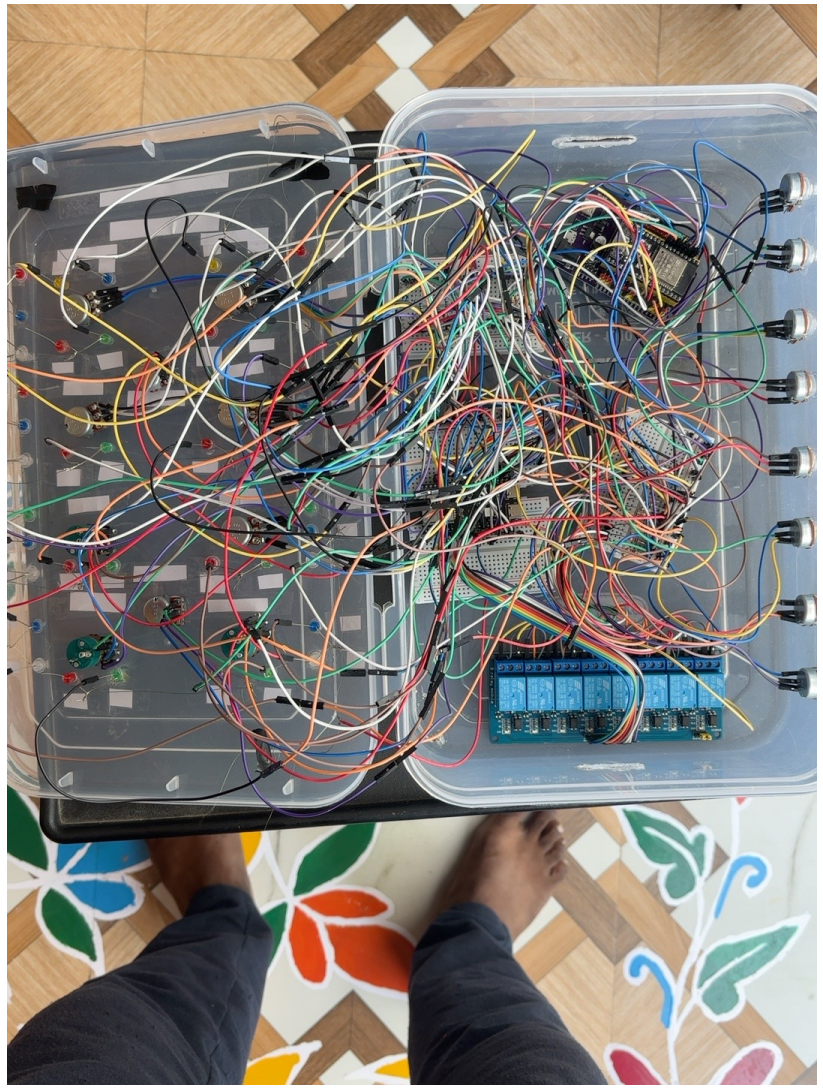
MQTT layer

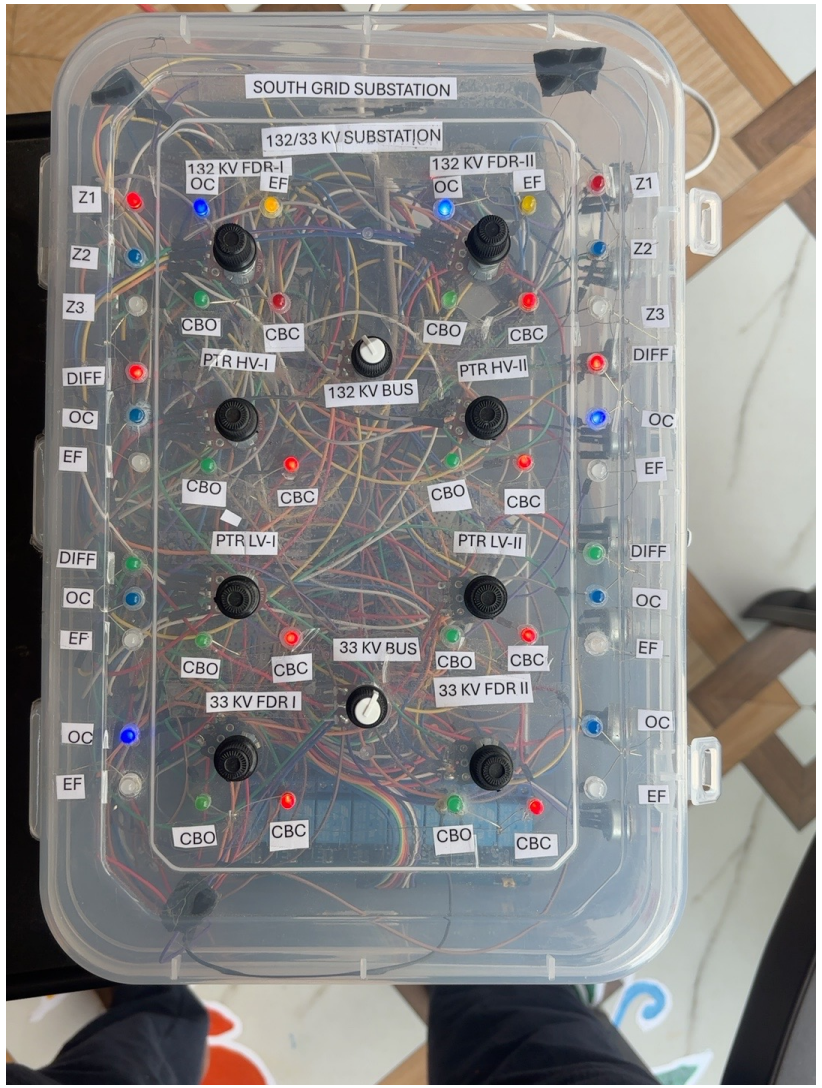
Wi-Fi + WiFiClientSecure + PubSubClient
HiveMQ Cloud / port 8883

EF input → **pickup** → **annunciation** → **UART / MQTT** → **event monitoring**

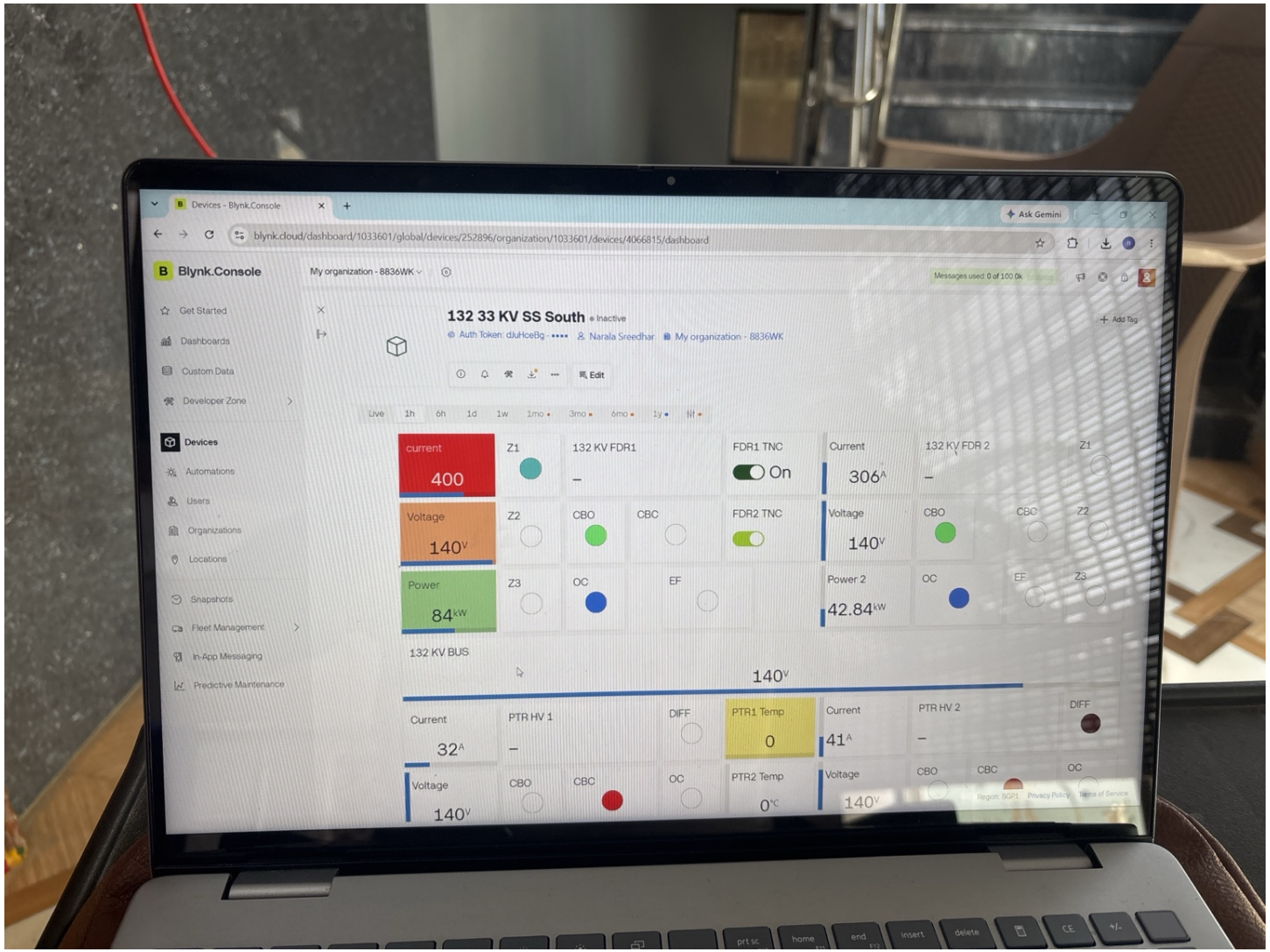
Blynk Supervisory Dashboard

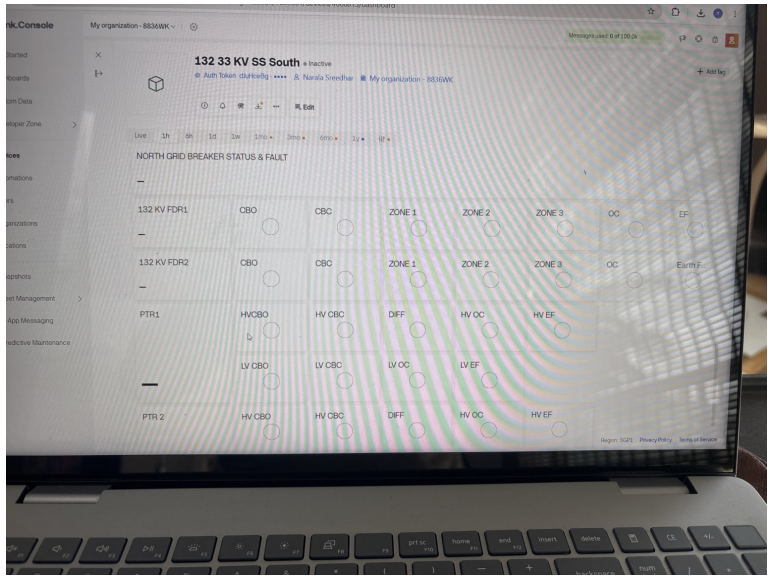
Prototype HMI view for measurements, breaker controls and protection status



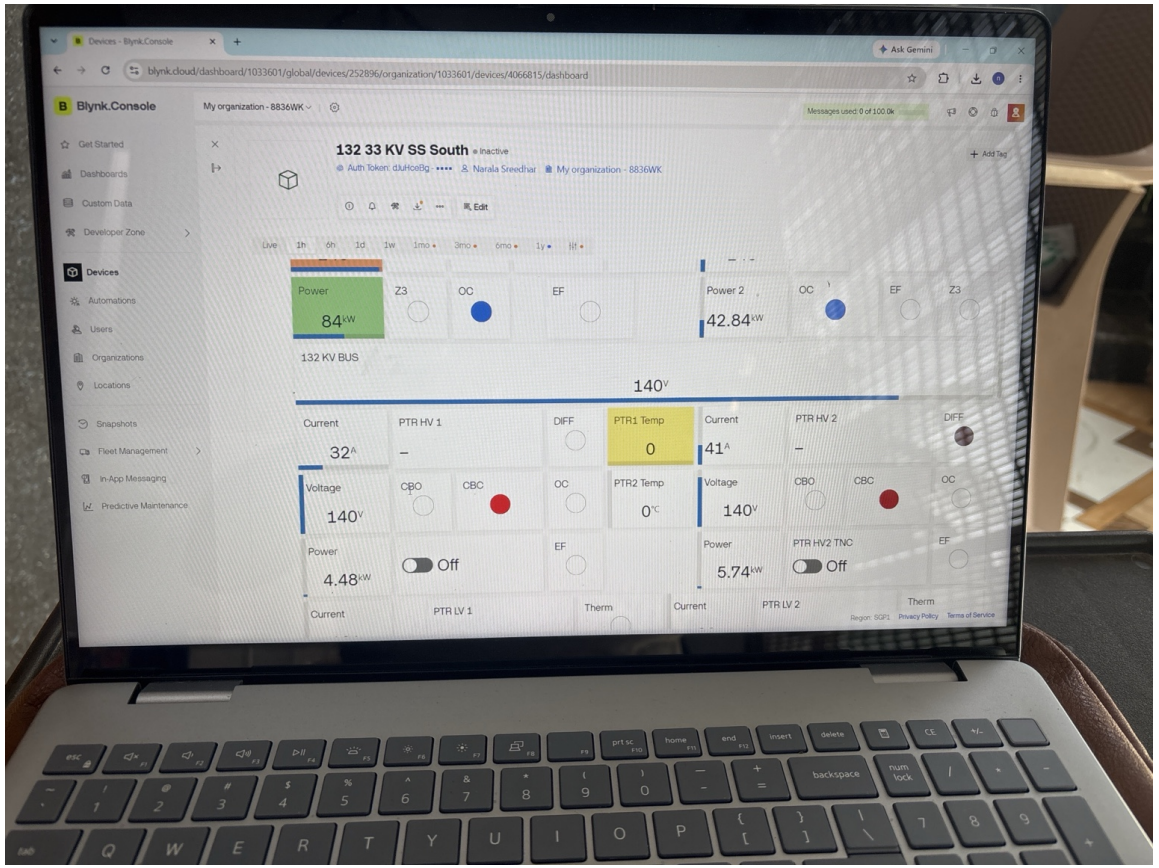


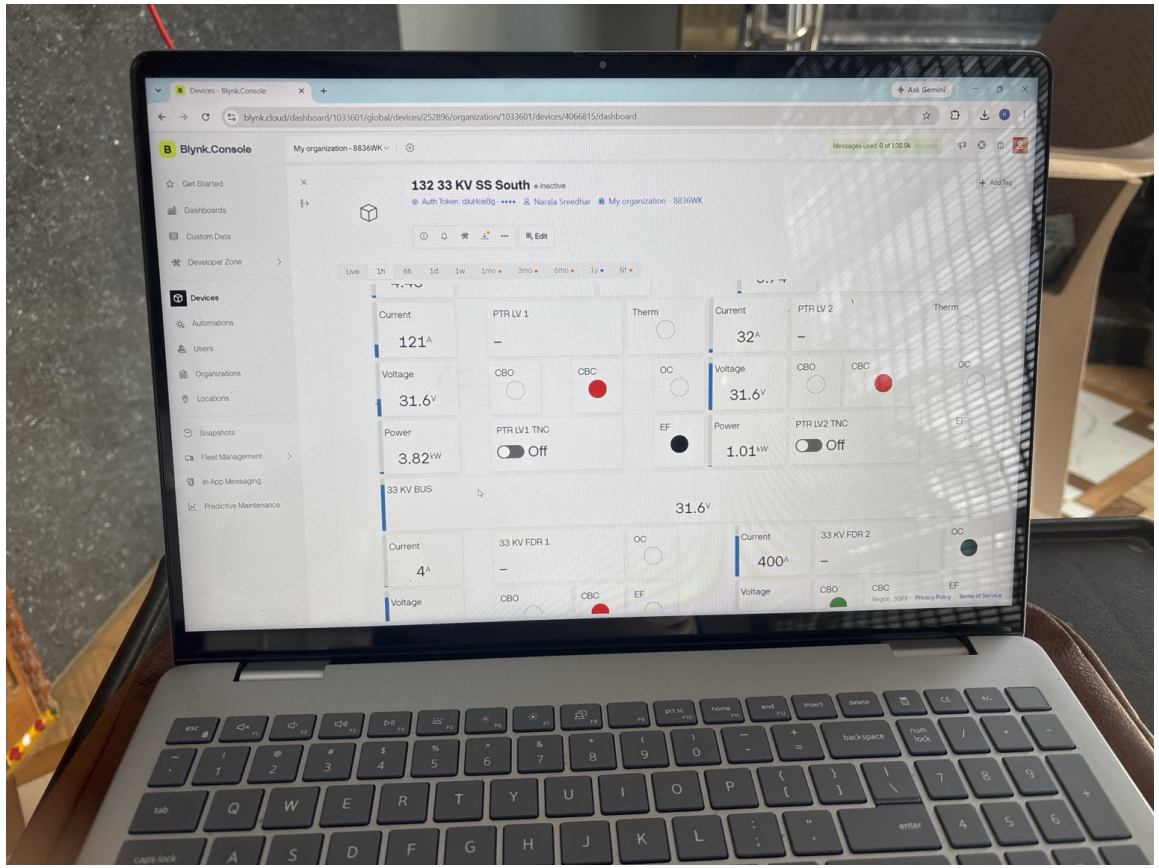
Protection & Breaker Status Demonstration





Physical Indications & Simulated Protection States





Software Execution Flow

From acquisition to protection, annunciation and supervisory communication

1. **ADC acquisition**
2. **Engineering conversion**
3. **Protection logic**
4. **Relay / annunciation**
5. **UART event**
6. **Blynk / MQTT**

Control path

Blynk virtual-pin commands operate prototype relay outputs. The source explicitly treats this as demonstration control requiring interlocks in practical systems.

Engineering Scope & Safety Notes

Prototype only

Educational low-voltage prototype; not an operational substation control system.

Isolation

Use isolated low-voltage hardware only.

Protection validation

Thresholds, scaling, timing and coordination require engineering validation.

Cybersecurity

Prototype uses setInsecure(); production requires certificate validation.

Control interlocking

Real breaker control needs permissives, interlocks and local/remote selection.

GitHub Project Package

README.md

firmware/ 2 Arduino files

docs/ engineering notes

images/ prototype photos

presentation/ PPTX + PDF

Standalone 132/33 kV Digital Substation Automation prototype

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